# ARON CHACKO

**ACADEMICS & RESEARCH HEAD | ROBOTICS & AI | APPLIED R&D | PHYSICAL AI**

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## PROFESSIONAL SUMMARY

Robotics and Artificial Intelligence professional, specializing in applied research and development, academic leadership, intelligent robotic systems, and technology-driven education. Currently serving as Academics and Research Head at Inker Robotics, leading academic programme development, technical training, applied R&D initiatives, project mentoring, and industry–academia engagement. Combines hands-on engineering experience with curriculum architecture and the delivery of practical technology programme across AI, computer vision, embedded systems, IoT, and robotics. Professional interests include Physical AI, embodied intelligence, autonomous systems, multimodal AI, edge intelligence, and human–robot interaction.

## CORE COMPETENCIES

- Applied Research & Development
- Academic Strategy & Curriculum Design
- Robotics & Intelligent Systems
- Artificial Intelligence & Computer Vision
- Generative and Multimodal AI
- Embedded Systems & IoT
- Edge AI
- Technical Programme Leadership
- Student and Team Mentoring
- Industry–Academia Collaboration

## PROFESSIONAL EXPERIENCE

**Academics and Research Head | Inker Robotic Solutions Pvt. Ltd. (Inker Robotics)**

Thrissur, Kerala, India | Present

- Lead development of academic programme, add-on courses, technical workshops, internships, and faculty development initiatives across AI, robotics, IoT, and embedded systems.
- Design industry-oriented curricula integrating technical foundations, hands-on hardware activities, software development, project-based learning, and assessment frameworks.
- Guide applied AI and robotics initiatives through technical exploration, prototyping, system integration, and demonstration, including computer vision, intelligent perception, embedded computing, and sensor integration.
- Develop interdisciplinary programme proposals connecting AI and robotics with computer science, electronics, biotechnology, and applied engineering.
- Mentor students, interns, educators, and engineering teams in Python, computer vision, machine learning, robotics, and embedded systems.
- Coordinate technology demonstrations, robotics exhibitions, STEM activities, and interactive AI experiences; contribute to institutional engagement and programme planning.

**Robotics Engineer | Inker Robotics**
Thrissur, Kerala, India | 2023-2026

- Programmed, configured, tested, and demonstrated robotic systems for educational, service, hospitality, and interactive applications.
- Developed AI demonstrations using Python, OpenCV, MediaPipe, object detection, and machine learning techniques.
- Worked with robotic arms, service robots, quadruped platforms, embedded controllers, and sensor-integrated systems.
- Supported hardware–software integration, technical troubleshooting, system validation, workshops, internships, and hands-on engineering sessions.

**Earlier Professional Experience**
**Python Developer** — Inmakes | **Technical Support Analyst** — CTDI | **Project Engineer** — Airon Telecom Services

Built experience in Python development, hardware diagnostics, technical troubleshooting, engineering project execution, and technical coordination. Add verified dates before submission.

## SELECTED R&D PROJECTS

**Autonomous Inventory Management Robot**
Explored an intelligent inventory system combining computer vision, depth sensing, and embedded computing. Technologies and components included YOLOv8n, LiDAR, Intel RealSense D455, Jetson Nano, Raspberry Pi 4, Arduino, Python, and serial communication. Focus: object identification and inventory monitoring.

**AI-Driven Solar Panel Optimization**
Research concept for identifying partial-shading patterns and supporting intelligent solar monitoring through computer vision. Explored OpenCV, TensorFlow, Roboflow, drone imagery, and edge-computing approaches.

**Generative AI Content Automation**
Explored an LLM-driven content-generation workflow using LangGraph and GPT-4o-mini, focusing on graph-based orchestration and structured task automation.

**Conversational AI and Interactive Robotics**
Explored integration of conversational intelligence with interactive robotic platforms for potential education, hospitality, demonstration, and service applications.

**Computer Vision and Interactive AI Applications**
Contributed to vision-based monitoring concepts, robotic photobooth experiences, AI-enabled interactive demonstrations, and digital-twin-related technology showcases. Portfolio includes a mix of implemented systems, prototypes, and exploratory concepts.

## ACADEMIC LEADERSHIP & PROGRAMME DEVELOPMENT

- Developed a Physical AI add-on course framework for BCA students covering processors, microcontrollers, ESP32, sensors, Python, computer vision, machine learning, Vision AI, and hardware–software integration.
- Designed IoT and embedded systems programme frameworks using ESP32, sensors, mobile applications, web interfaces, dashboards, and Python-based computer vision applications.

- Prepared interdisciplinary AI and robotics programme proposals for engineering and biotechnology education, including computational biology, AlphaFold Server, AI research tools, prompt engineering, biomechanics, and bio-robotics.
- Structured programme outcomes, practical exercises, project-based learning activities, and assessments for school, college, and faculty audiences.

## TRAINING, MENTORSHIP & IMPACT

Previously discussed profile estimates, to be verified before external use:

- 1,000+ hours of training and workshop delivery.
- 500+ students mentored across technical learning and project-oriented programmes.
- 50+ workshops and training programmes across AI, robotics, Python, computer vision, and related technologies.
- Engagement with 10+ educational institutions through training, demonstrations, or technology programmes.
- Facilitated talks and awareness sessions for professionals covering AI tools, ethical cosiderations, and bias.

## SPEAKING & RECOGNITION

- Speaker, AL EVOLVE Meetup, Technopark, Thiruvananthapuram (2026): Technical Talk on Physical AI and Robotics.
- Faculty Development Programmes: contributed talk content on Text and Image Generation Mechanics and Multimodal Generative AI.
- Industry Recognition: Recognized by various organizations and educational institutions for delivering expert-led sessions, technical talks, and workshops in AI, robotics, and emerging technologies.
- Professional event exposure includes GenAI Conclave, Huddle Global, and robotics and AI technology showcases.

## TECHNICAL SKILLS

**Programming**: Python

**AI & Computer Vision**: OpenCV, MediaPipe, YOLOv8, TensorFlow, object detection, machine learning, Generative AI, LLM applications, multimodal AI, prompt engineering, RAG Pipline

**Web Development**: LangGraph, GPT-based applications, FastAPI, Flask, Django

**Robotics & Embedded Systems**: ROS, Arduino, ESP32, Raspberry Pi, Jetson Nano, robotic arm programming, sensors, serial communication, embedded computing

**Perception & Hardware**: LiDAR, Intel RealSense depth cameras, camera-based perception, edge AI, hardware–software integration

**Leadership**: Curriculum development, programme design, applied R&D, project mentoring, technical training, research coordination, institutional collaboration

## EDUCATION

**Bachelor of Technology (B.Tech.) in Electronics and Communication Engineering**
University of Calicut, Kerala, India | 2018